

SOME EXTENSIONS OF THE WISHART MOMENT GENERATING FUNCTION

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1. Introduction

Consider the moment generating function (m.g.f.) of the central Wishart distribution with parameter matrix Σ : p x p (Anderson (1958)),

$$(1.1) \quad M(T) = |I - 2\Sigma T|^{-\frac{1}{2}n} \\ = \sum_{k=0}^{\infty} \sum_K \binom{\frac{1}{2}n}{K} \frac{C_K(2\Sigma T)}{k!} \quad (\text{James (1964)})$$

where $T = [t_{ij}]$ is a p x p symmetric matrix. In the special case when $\Sigma = \sigma^2 I_p$ and $HTH' = \begin{pmatrix} t_1 & & 0 \\ & \ddots & \\ 0 & & t_p \end{pmatrix}$ the m.g.f. is a function of the t_i . For general Σ the m.g.f. is no longer

an explicit function of the t_i .

We symmetrise the zonal polynomial in (1.1) (Greenacre (1973)), by which we mean that $C_K(2\Sigma T)$ is replaced by its average

$$\int_{O(p)} C_K(2\Sigma HTH') dH$$

with respect to the invariant measure dH , on the orthogonal group $O(p)$ of p x p orthogonal matrices H . The invariant measure is normalised so that the total measure of $O(p)$ is unity. The m.g.f. (1.1) is not invariant under congruence transformation by the orthogonal group $O(p)$, $T \rightarrow HTH'$, $H \in O(p)$, thus the symmetrised m.g.f. is

$$(1.2) \quad M_s(T) = \sum_{k=0}^{\infty} \sum_K \binom{\frac{1}{2}n}{K} \frac{C_K(2\Sigma)C_K(T)}{C_K(I_p)k!}.$$

The m.g.f. (1.2) is now also a function of the t_i . For $\Sigma = \sigma^2 I_p$ we find that

coefficient $t_{ij} = 0$,

coefficient $t_{ii} = t_{jj}$,

coefficient $(t_{ij})^2 = (t_{lm})^2$ etc.

The purpose of this note is to introduce a new class of m.g.f.'s, with desirable properties, that are in certain cases equivalent to the actual m.g.f. and may in fact be used in its place.

The method of m.g.f.'s is useful in the study of distribution problems in multivariate populations. The distributions of some statistics can be derived by using the inverse Laplace transform. We mention here only as examples the Wishart and noncentral Wishart m.g.f.'s.

2. Definitions and properties

The following results are of importance.

Definition 2.1 (Greenacre (1973))

The symmetrised (multivariate) density of the positive definite symmetric matrix $A: p \times p$ is defined as

$$f_s(A) = \int_{O(p)} f(HAH') dH$$

where A has the density function $f(A)$.

Definition 2.2

The symmetrised m.g.f. of the positive definite symmetric matrix $A: p \times p$ is defined as

$$M_s(T) = \int_{O(p)} M(HTH') dH$$

where $M(T)$ is the ordinary m.g.f. of the density function $f(A)$.

The symmetrised m.g.f. satisfies the conditions of a m.g.f. and has the following important property.

Theorem 2.1

The symmetrised m.g.f. $M_s(T)$ is the m.g.f. of the symmetrised density function $f_s(A)$.

Proof The m.g.f. of $f_s(A)$ is defined by

$$M(T) = \int_{A > 0} e^{\text{tr}TA} f_s(A) dA$$

$$= \int_{A \succ 0} e^{\text{tr}TA} \int_{O(p)} f(HAH') dH dA \quad (\text{def. 2.1}).$$

By changing the order of integration and transforming $A \rightarrow H'AH$, $H \in O(p)$, with Jacobian equal to $|H|^{p+1} = 1$ (Deemer and Olkin (1959)), we obtain

$$\begin{aligned} M(T) &= \int_{O(p)} \int_{A \succ 0} e^{\text{tr}HTH'} A \cdot f(A) dA dH \\ &= \int_{O(p)} M(HTH') dH \\ &= M_s(T) \quad (\text{def. 2.2}). \end{aligned}$$

The symmetrised density has the property that the distribution of the characteristic roots of A is the same for $f_s(A)$ and $f(A)$ (Greenacre (1973)).

3. Extensions of moment generating functions

Consider the following examples.

Example 3.1

The m.g.f. for the noncentral Wishart matrix $A: p \times p$ is given by Herz (1955) as

$$\begin{aligned} (3.1) \quad M(T) &= E[e^{\text{tr}AT}] \\ &= |I - 2\Sigma T|^{-\frac{1}{2}n} e^{\text{tr}2\Omega T \Sigma (I - 2\Sigma T)^{-1}} \end{aligned}$$

where $\Omega = \frac{1}{2}\Sigma^{-1}\mu\mu'$ is the matrix of noncentrality parameters.

Making the transformation $T \rightarrow HTH'$ in (3.1) and integrating over $H \in O(p)$ we have

$$M_s(T) = \int_{O(p)} |I - 2\Sigma HTH'|^{-\frac{1}{2}n} e^{\text{tr}2\Omega HTH' \Sigma (I - 2\Sigma HTH')^{-1}} dH$$

Hence the symmetrised m.g.f. of $\text{tr } A$ is

$$\begin{aligned} M_s(t) &= |I - 2\Sigma t|^{-\frac{1}{2}n} e^{\text{tr} \Omega 2\Sigma t (I - 2\Sigma t)^{-1}} \\ &= M(t) \end{aligned}$$

which is the ordinary m.g.f. of $\text{tr } A$.

Example 3.2

The Laplace transform of the central Wishart distribution is defined by

$$g(T) = E[e^{-\text{tr } AT}]$$

$$= |I + 2\Sigma T|^{-\frac{1}{2}n}$$

$$= |2\Sigma T|^{-\frac{1}{2}n} |I + \frac{1}{2}\Sigma^{-1}T^{-1}|^{-\frac{1}{2}n}.$$

The symmetrised Wishart distribution will be defined by inverting the symmetrised Laplace transform

$$g_s(T) = |2\Sigma|^{-\frac{1}{2}n} |T|^{-\frac{1}{2}n} \sum_{k=0}^{\infty} \sum_K \binom{\frac{1}{2}n}{K} \frac{C_K(-\frac{1}{2}\Sigma^{-1}) C_K(T^{-1})}{C_K(I_p) k!}$$

of its density function.

Since

$$\frac{2^{\frac{1}{2}p(p-1)}}{(2\pi i)^{\frac{1}{2}p(p+1)}} \int_{R(T) > 0} e^{\text{tr } AT} |T|^{-\frac{1}{2}n} C_K(T^{-1}) dT = \frac{1}{\Gamma_p(\frac{1}{2}n, K)} |A|^{\frac{1}{2}(n-p-1)} C_K(A),$$

we have

$$f_s(A) = \frac{|2\Sigma|^{-\frac{1}{2}n}}{\Gamma_p(\frac{1}{2}n)} |A|^{\frac{1}{2}(n-p-1)} {}_0F_0(-\frac{1}{2}\Sigma^{-1}; A).$$

The distribution of the roots of A can now be obtained by using the theorem of Anderson (1958).

Using the approximation for the noncentral Wishart by a central Wishart with Σ replaced by $\Sigma^* = \Sigma + \frac{1}{n} \sum_{\alpha=1}^n \mu_\alpha \mu'_\alpha$ (Steyn and Roux (1972)), we can find a symmetrised approximated noncentral Wishart distribution.

Example 3.3

A m.g.f. for the noncentral Wishart matrix $A: p \times p$ when sampling from a multivariate normal population with $\Sigma = \sigma^2 I_p$ is

$$(3.2) \quad M^*(T) = |I - 2\sigma^2 T|^{-\frac{1}{2}n} e^{\text{tr } 2\sigma^2 \Omega T (I - 2\sigma^2 T)^{-1}}$$

$$= |I - 2\sigma^2 T|^{-\frac{1}{2}n} \sum_{k=0}^{\infty} \sum_K \frac{C_K(2\sigma^2 \Omega T (I - 2\sigma^2 T)^{-1})}{k!} \quad (\text{James (1964)}).$$

Making the transformation $T \rightarrow HTH'$ in (3.2) and integrating over $H \in O(p)$, using

$$\int_{O(p)} C_K(2\sigma^2 \Omega H T (I - 2\sigma^2 T)^{-1} H') dH = \frac{C_K(\Omega) C_K(2\sigma^2 T (I - 2\sigma^2 T)^{-1})}{C_K(I_p)}, \text{ then}$$

$$(3.3) \quad M_s^*(T) = |I - 2\sigma^2 T|^{-\frac{1}{2}n} \sum_{k=0}^{\infty} \sum_K \frac{C_K(\Omega) C_K(2\sigma^2 T (I - 2\sigma^2 T)^{-1})}{C_K(I_p) k!}.$$

This is a special case of the m.g.f. defined by Roux and Raath (1973). Constantine (1966) has derived that

$$M_s^*(T) = \sum_{k=0}^{\infty} \sum_K L_K^{\frac{1}{2}(n-p-1)} (-\Omega) \frac{C_K(2\sigma^2 T)}{C_K(I_p)k!},$$

the generating function for the Laguerre polynomials.

It follows from (3.3) and Roux and Raath (1973) that the symmetrised noncentral distribution is given by

$$f_s^*(A) = f_s(A) \sum_{k=0}^{\infty} \sum_K \frac{1}{(\frac{1}{2}n)_K} \frac{C_K(-\Omega)}{C_K(I_p)k!} L_K^{\frac{1}{2}(n-p-1)} \left(\frac{1}{2\sigma^2} A \right)$$

where

$$f_s(A) = f(A) = \frac{(2\sigma^2)^{-\frac{1}{2}pn}}{\Gamma_p(\frac{1}{2}n)} e^{-\frac{1}{2\sigma^2} \text{tr} A} |A|^{\frac{1}{2}(n-p-1)}, \quad A: p \times p > 0.$$

Thus $f_s^*(A)$ is in the form of a symmetrised central Wishart density with $\Sigma = \sigma^2 I_p$ multiplied by a weighting function, and the distribution of the roots of A can again be obtained as an immediate consequence of Anderson (1958).

Putting $\frac{1}{2\sigma^2} I - T = W$ in (3.2) and using the fact that

$$\text{tr } 2\sigma^2 \Omega T (I - 2\sigma^2 T)^{-1} = \text{tr } \{ \Omega - \Omega(I - 2\sigma^2 T)^{-1} \},$$

the symmetrised m.g.f. follows as

$$M_s^* \left(W + \frac{1}{2\sigma^2} I \right) = (2\sigma^2)^{-\frac{1}{2}np} |W|^{-\frac{1}{2}n} e^{-\text{tr} \Omega} \sum_{k=0}^{\infty} \sum_K \frac{C_K(\frac{1}{2\sigma^2} \Omega) C_K(W^{-1})}{C_K(I_p)k!}$$

where $\Omega = \frac{1}{2\sigma^2} \mu \mu'$.

Applying the Laplace inversion formula we obtain, after algebraic manipulation, the symmetrised noncentral Wishart distribution in the form

$$f_s^*(A) = f_s(A) e^{-\text{tr} \Omega} \sum_{k=0}^{\infty} \sum_K \frac{C_K(\frac{1}{2\sigma^2} \Omega) C_K(A)}{(\frac{1}{2}n)_K C_K(I_p)k!}.$$

This distribution has been derived by Greenacre (1972).

The following is an example dealing with the division of the matrix A.

Example 3.4

From (1.1) it follows that

$$(3.4) \quad M(T_1, T_2) = E[e^{\text{tr}(X_1 T_1 + X_2 T_2)}] \\ = |I - 2T_1|^{-\frac{1}{2}n} |I - 2T_2|^{-\frac{1}{2}n} |I - 2\Sigma_{21} T_1 (I - 2T_1)^{-1} 2\Sigma_{12} T_2 (I - 2T_2)^{-1}|^{-\frac{1}{2}n}$$

is a possible m.g.f. for a two-matrix multivariate distribution in which each of the random matrices has a probability density function

$$f(A_i) = \frac{2^{-\frac{1}{2}np}}{\Gamma_p(\frac{1}{2}n)} e^{-\frac{1}{2}\text{tr}A_i} |A_i|^{\frac{1}{2}(n-p-1)} \text{ for } A_i: p \times p > 0.$$

Moreover the vectors in the parent normal distribution are positively correlated with symmetric correlation matrix $\Sigma_{12} = \Sigma'_{21}$.

Making the transformations $T_1 \rightarrow H_1 T_1 H'_1$ and $T_2 \rightarrow H_2 T_2 H'_2$ in (3.4) and integrating over $H_1 \in O(p)$ and $H_2 \in O(p)$,

$$(3.5) \quad M_s(T_1, T_2) = \sum_{k=0}^{\infty} \sum_K \left(\frac{1}{2}n\right)_K \frac{C_K(\Sigma_{12}\Sigma_{21})}{k! [C_K(I_p)]^2} |I - 2T_1|^{-\frac{1}{2}n} C_K(2T_1(I - 2T_1)^{-1}) \\ \cdot |I - 2T_2|^{-\frac{1}{2}n} C_K(2T_2(I - 2T_2)^{-1})$$

as has been defined by Roux and Raath (1973).

It follows from (3.5) and Roux and Raath (1973) that

$$f_s(A_1, A_2) = f_s(A_1) f_s(A_2) \left[\sum_{k=0}^{\infty} \sum_K \frac{1}{K! \left(\frac{1}{2}n\right)_K} \frac{C_K(\Sigma_{12}\Sigma_{21})}{k! [C_K(I_p)]^2} L_K^{\frac{1}{2}(n-p-1)} \left(\frac{1}{2}A_1\right) \right. \\ \left. \cdot L_K^{\frac{1}{2}(n-p-1)} \left(\frac{1}{2}A_2\right) \right].$$

If we take, instead of the m.g.f. in (3.4), the more general m.g.f. with parameters n_1, n_2 and n ,

$$M(T_1, T_2) = |I - 2T_1|^{-\frac{1}{2}n_1} |I - 2T_2|^{-\frac{1}{2}n_2} |I - 2\Sigma_{21} T_1 (I - 2T_1)^{-1} 2\Sigma_{12} T_2 (I - 2T_2)^{-1}|^{-\frac{1}{2}n}$$

and repeat the procedure, we obtain the following series

$$f_s^{n_1}(A_1)f_s^{n_2}(A_2) \left[\sum_{k=0}^{\infty} \sum_K \left(\frac{1}{2}n \right)_K \frac{C_K(\Sigma_1 \Sigma_2)}{C_K(I_p)^k!} \frac{1}{\left(\frac{1}{2}n_1 \right)_K \left(\frac{1}{2}n_2 \right)_K} \right. \\ \left. \cdot L_K^{\frac{1}{2}(n_1-p-1)} \left(\frac{1}{2}A_1 \right) L_K^{\frac{1}{2}(n_2-p-1)} \left(\frac{1}{2}A_2 \right) \right]$$

where

$$f_s^{n_i}(A_i) = 2^{-\frac{1}{2}n_i p} \frac{1}{\Gamma_p\left(\frac{1}{2}n_i\right)} e^{-\frac{1}{2}\text{tr}A_i} |A_i|^{-\frac{1}{2}(n_i-p-1)} \quad \text{for } i=1,2.$$

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